Design and Analysis of a Lightweight Post Quantum Encryption Model for IoT Devices Using CRYSTALS Kyber Algorithm

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***Abstract*— The rapid expansion of the Internet of Things (IoT) has intensified the need for cryptographic mechanisms that remain secure against quantum threats while remaining feasible for resource-constrained devices. Classical algorithms such as RSA and ECC are vulnerable to quantum attacks, motivating the adoption of lattice-based Post-Quantum Cryptography (PQC) algorithms like CRYSTALS-Kyber. This study presents the design and analysis of a lightweight post-quantum encryption model tailored for IoT environments. The proposed framework integrates an optimized CRYSTALS-Kyber Key Encapsulation Mechanism (KEM) within low-power communication protocols such as MQTT and wireless sensor networks. The model leverages architectural simplifications, physical-layer adaptations, and modular arithmetic optimizations to reduce computational overhead and energy consumption. Through literature synthesis and performance analysis, Kyber demonstrates a superior balance between security, efficiency, and scalability compared to other NIST-standardized PQC schemes. The findings highlight that, with appropriate hardware and algorithmic tuning, CRYSTALS-Kyber can serve as a practical foundation for lightweight, quantum-resilient encryption in next-generation IoT systems.**

Keywords*— *CRYSTALS-Kyber; Post-Quantum Cryptography; Internet of Things; Lightweight Encryption; Lattice-Based Cryptography; Key Encapsulation Mechanism; Resource-Constrained Devices; MQTT Protocol; Wireless Communication; Quantum-Resilient Security

I. INTRODUCTION

The rapid proliferation of the Internet of Things (IoT) has led to the interconnection of billions of heterogeneous devices, enabling smarter environments across healthcare, industry, and transportation. However, this massive connectivity has also expanded the attack surface, exposing IoT systems to severe vulnerabilities. Most IoT devices are resource-constrained, operating with limited computational power, memory, and energy capacity. These constraints make it difficult to adopt conventional encryption schemes that rely on heavy mathematical operations. At the same time, advances in quantum computing are projected to break widely used cryptographic standards such as RSA and Elliptic Curve Cryptography (ECC), posing a significant threat to the long-term security of IoT ecosystems [2], [6]. Consequently, the design of lightweight Post-Quantum Cryptography (PQC) has become an urgent research frontier to secure resource-limited networks against quantum-enabled adversaries.

Among the PQC schemes standardized by the National Institute of Standards and Technology (NIST) in 2024, CRYSTALS-Kyber has emerged as a leading Key Encapsulation Mechanism (KEM) due to its efficiency, compact key sizes, and strong resistance to quantum attacks [1]. Kyber is based on the Module Learning with Errors (MLWE) problem, which remains computationally infeasible even for quantum computers. Several studies have explored its adaptability in IoT contexts. González de la Torre et al. [1] proposed a wireless-based Kyber adaptation at the physical layer using Additive White Gaussian Noise (AWGN) modeling, demonstrating low key error probability while maintaining cryptographic robustness. Their work represents an innovative attempt to reduce protocol-layer complexity by embedding post-quantum security directly into communication channels. In a complementary direction, Kim and Seo [3] introduced an optimized KEM-MQTT protocol implementing Kyber-512 on 8-bit AVR microcontrollers, achieving up to 85% performance improvement in key encapsulation and decapsulation operations. Such efforts highlight the potential of Kyber to deliver lightweight yet secure key exchange mechanisms in embedded IoT systems.

Despite these advancements, several challenges persist in deploying PQC efficiently within IoT environments. Mahdi and Abdullah [2] emphasized that most post-quantum algorithms still demand high computational resources, leading to increased latency and energy consumption on constrained devices. Their review on lightweight PQC highlighted the need for algorithmic refinement, hardware acceleration, and hybrid cryptographic models to ensure practicality. Moreover, physical-layer adaptations such as WKyber [1] remain primarily at simulation level, with limited integration into real-world IoT protocols. Current studies tend to optimize isolated aspects—either algorithmic speed or hardware efficiency—without offering a unified lightweight model that systematically balances security, scalability, and performance under IoT constraints.

The existing research gap thus lies in the lack of an integrated lightweight encryption framework that combines algorithmic optimization with protocol-level adaptation for post-quantum IoT systems. There is also limited comparative analysis between the various implementation strategies of Kyber—such as hardware-assisted computation, MQTT protocol integration, and physical-layer embedding—under consistent performance metrics. Addressing this gap is crucial for achieving quantum-resilient communication that remains deployable on low-end IoT hardware.

Therefore, this research focuses on the design and analysis of a lightweight post-quantum encryption model based on the CRYSTALS-Kyber algorithm for IoT devices. The study aims to synthesize findings from recent PQC developments, evaluate performance trade-offs among lightweight Kyber implementations, and propose a conceptual model for efficient key encapsulation in resource-constrained environments. The expected outcome is a framework that maintains post-quantum security guarantees while optimizing computational and energy efficiency, providing a feasible foundation for secure IoT ecosystems in the quantum era.

II. METHOD

This study adopts a systematic literature review approach to identify, analyze, and synthesize recent research related to the implementation of Post-Quantum Cryptography (PQC)—specifically the CRYSTALS-Kyber algorithm—in resource-constrained IoT environments. The objective of this review is to evaluate the progress of lightweight encryption models that integrate Kyber into low-power communication systems, such as wireless sensor networks and MQTT-based IoT protocols.

A. Literature Search Strategy

The literature search was conducted using the Scopus database, recognized for its comprehensive coverage of peer-reviewed scientific publications. The search focused on the Article title, Abstract, and Keywords fields to ensure relevance. The following query string was used:

"CRYSTALS Kyber" AND "Internet of Things" AND "Post Quantum Cryptography"

This query yielded 27 documents from 2020 to 2025. Additional filters were applied to refine the results:

- Language: English

- Subject areas: Computer Science, Engineering

- Document type: Articles and conference papers

- Publication period: 2020–2025

B. Selection Criteria

From the initial 27 results, 15 papers were selected based on their direct relevance to the integration of post-quantum algorithms with IoT devices. The inclusion criteria were:

Criterion Type	***Description***
Inclusion	
Algorithm relevance	Must involve CRYSTALS-Kyber or other lattice-based PQC algorithms applicable to IoT or embedded systems
Lightweight adaptation	Must discuss optimization for resource-constrained devices, including computational, memory, or energy efficiency
Implementation focus	Must describe practical implementation or simulation, e.g., on microcontrollers, FPGA, or IoT testbeds
Quantitative evaluation	Must report performance metrics such as latency, key generation time, energy use, or memory footprint
Publication period	Primarily 2020–2025; earlier foundational works included if directly cited by NIST or IACR standards
Publication type	Peer-reviewed journals, conferences (IEEE, ACM, Elsevier, MDPI, IACR)
Accessibility	Full-text available via institutional or open-access sources
Exclusion	
Theoretical only	Papers focusing solely on PQC mathematics without IoT or implementation context
Non-lattice PQC	Works unrelated to lattice-based or Kyber-style cryptography (e.g., hash-based or code-based PQC)
Lack of metrics	Studies that do not provide measurable performance evaluation
Out of scope	Research addressing unrelated applications such as QKD or quantum networking protocols
Language	Available in English
Non-peer-reviewed	Excluded articles, blogs, or informal white papers not indexed in Scopus

C. Data Extraction

After the selection process, data from the 18 chosen papers were extracted using a structured template to ensure consistency and comparability. Each paper was reviewed in detail to capture essential elements related to algorithmic implementation, optimization techniques, and performance outcomes of CRYSTALS-Kyber and other lightweight PQC schemes in IoT contexts.

Category	***Extracted Information***
Bibliographic details	author(s), publication year, source type, and DOI.
Objective and contribution	Application domain, interaction type, session duration
Data	main goal of the study and proposed model or framework.
Implementation environment	type of IoT hardware, communication protocol, or platform used (e.g., 8-bit AVR, RISC-V, MQTT, wireless channel, FPGA).
Optimization strategy	techniques applied for resource efficiency, such as modular arithmetic simplification, hardware acceleration, or parameter tuning (Kyber512, Kyber768).
Performance metrics	measured results including key generation time, encryption/decryption latency, memory footprint, and energy consumption.
Evaluation approach	experimental setup, simulation tools, or benchmark datasets.
Research outcomes	key findings on performance improvement, scalability, or trade-offs.
Identified gaps	unresolved limitations, such as integration challenges, energy efficiency, or scalability in real-world IoT deployments.

III. RESULTS AND ANALYSIS

This section presents the results of the literature review and analytical synthesis based on fifteen selected studies focusing on lightweight post-quantum cryptography for Internet of Things (IoT) environments, with an emphasis on the CRYSTALS-Kyber algorithm. The selected works, published between 2021 and 2025, represent current advancements in algorithmic optimization, protocol-level integration, and hardware co-design approaches aimed at enhancing quantum-resistant encryption within resource-constrained IoT devices. Through comparative examination, the findings highlight emerging trends in reducing computational complexity, improving encryption speed, and balancing energy efficiency without compromising security. The analysis also identifies persistent challenges related to scalability, standardization, and real-world implementation, which form the foundation for designing a unified lightweight post-quantum encryption model suitable for next-generation IoT systems.

A. Overview of Selected Studies

From the total of twenty-seven Scopus-indexed publications retrieved using the search query "CRYSTALS-Kyber" AND "Internet of Things" AND "Post-Quantum Cryptography," fifteen papers were selected based on the inclusion criteria described in the methodology section. These papers encompass a diverse range of approaches that collectively reflect the state of the art in post-quantum security for IoT ecosystems. Most studies were published between 2023 and 2025, indicating the rapid and ongoing interest in lightweight implementations of post-quantum cryptographic algorithms following the standardization of CRYSTALS-Kyber by the National Institute of Standards and Technology (NIST) in 2024.

The reviewed papers can be categorized into three major themes: (1) algorithmic and mathematical optimization of CRYSTALS-Kyber to enhance performance on constrained devices, (2) integration of post-quantum encryption into IoT communication protocols such as MQTT, TLS, and wireless physical layers, and (3) hardware and FPGA-based co-design architectures targeting energy-efficient acceleration. Algorithmic optimization studies, including those by Huang et al. and Ye et al., focused on modular arithmetic simplification and parameter tuning for faster key generation and encapsulation. Protocol-level studies, such as those by Kim and Seo, Torre et al., and Paul et al., examined how Kyber can be integrated into IoT protocols to achieve secure yet lightweight communication. Meanwhile, hardware-focused research explored FPGA, RISC-V, and embedded architectures to accelerate Kyber operations and minimize latency while maintaining low power consumption.

Collectively, these studies demonstrate the increasing maturity of post-quantum cryptography in the IoT domain, where the primary challenge lies in balancing security, efficiency, and practicality. The analysis in subsequent sections synthesizes these findings to outline the evolution of CRYSTALS-Kyber implementation strategies and to identify existing research gaps that motivate the design of an adaptive lightweight encryption model for future IoT systems.

B. Algorithmic Optimization in Lightweight Post-Quantum Encryption

Algorithmic optimization represents one of the most active research directions in adapting post-quantum cryptography to resource-constrained IoT environments. Several studies within the reviewed literature focus on improving the computational performance of the CRYSTALS-Kyber algorithm without compromising its quantum-resistant security. Huang et al. (2024) introduced an enhanced Plantard arithmetic approach for 32-bit microcontrollers that significantly reduced modular multiplication costs, achieving up to a 25% improvement in runtime efficiency compared to standard Kyber implementations. Similarly, Ye et al. (2024) designed a highly efficient lattice-based cryptography processor for IoT applications, leveraging pipelined architectures and hardware parallelism to accelerate polynomial operations, key generation, and encapsulation processes.

Mahdi and Abdullah (2025) provided a comprehensive review of lightweight post-quantum cryptographic schemes, emphasizing the suitability of lattice-based algorithms such as CRYSTALS-Kyber and NTRU for low-power IoT networks. Their findings indicated that the Kyber512 variant offers the best trade-off between key size, encryption speed, and energy consumption, making it ideal for small-scale IoT deployments. Fitzgibbon and Ottaviani (2024) conducted performance benchmarking across various PQC algorithms, confirming that Kyber consistently outperformed other NIST finalists—such as Saber and BIKE—in terms of key generation and decapsulation latency on microcontrollers.

In addition, several studies explored algorithmic fine-tuning through arithmetic simplification and noise-sampling reduction. These approaches aimed to minimize the computational burden associated with the Learning With Errors (LWE) problem while maintaining security guarantees. Collectively, the algorithm-level analyses demonstrate that the computational cost of Kyber can be substantially reduced through optimized polynomial arithmetic, improved random number generation, and efficient modular compression functions. Such enhancements serve as the foundation for the proposed lightweight encryption framework, ensuring that post-quantum security remains achievable even on low-power IoT devices.

C. Protocol-Level Integration for IoT Security

Beyond algorithmic improvements, an essential dimension of post-quantum adaptation for IoT involves the integration of CRYSTALS-Kyber into existing communication protocols to ensure end-to-end quantum-resistant security. This direction addresses the challenge of embedding post-quantum encryption within lightweight communication frameworks such as MQTT, TLS, and CoAP—protocols that are widely used in IoT but limited by bandwidth, latency, and processing power constraints.

Kim and Seo (2025) introduced KEM-MQTT, a novel post-quantum variant of the Message Queuing Telemetry Transport protocol that integrates the Kyber Key Encapsulation Mechanism (KEM). Their implementation on 8-bit AVR sensor nodes demonstrated remarkable efficiency, reducing key generation and encapsulation times by more than 80% compared to baseline implementations, while maintaining full compatibility with TLS 1.3 security guarantees. This study proved that even microcontroller-based IoT nodes can achieve quantum-safe communication

when the cryptographic layer is tightly integrated with the transport protocol.

Similarly, Torre et al. (2025) proposed Wireless Kyber (WKyber), a physical-layer adaptation of the Kyber encryption mechanism that leverages Additive White Gaussian Noise (AWGN) channels and Bose–Chaudhuri–Hocquenghem (BCH) coding. This approach reduces reliance on upper-layer encryption, embedding quantum-safe key exchange directly into the modulation process. Simulation results revealed a low key error probability and stable performance, demonstrating that post-quantum encryption can coexist efficiently with wireless transmission characteristics.

Complementary to these studies, Paul et al. (2021) presented a TPM-based post-quantum TLS framework for IoT environments, employing Kyber within a hybrid mutual authentication scheme that combines traditional ECC and PQC primitives. Their results showed improved handshake performance and enhanced resilience against quantum-based key-recovery attacks. Likewise, Parameswarath et al. (2024) developed a quantum-safe authentication protocol tailored for transportation IoT systems, demonstrating that Kyber-based key encapsulation can secure real-time communication while maintaining acceptable latency levels.

Collectively, these works emphasize that protocol-level integration is among the most practical pathways toward scalable post-quantum IoT security. By aligning cryptographic primitives with lightweight communication stacks, researchers are enabling IoT devices to transition toward post-quantum readiness without substantial redesign of existing network architectures. However, interoperability and standardization challenges remain, particularly in ensuring compatibility across diverse IoT ecosystems and balancing the trade-off between cryptographic strength and operational efficiency.

D. Hardware and Co-Design Implementations

While software-level optimizations and protocol integration improve efficiency, the most significant performance gains in post-quantum cryptography (PQC) for IoT have been achieved through hardware acceleration and hardware/software co-design architectures. These approaches address the intrinsic limitations of constrained devices by offloading computationally intensive operations, such as polynomial multiplication and modular reduction, to dedicated hardware units.

Dinh et al. (2024) proposed a multi-mode CRYSTALS-Kyber accelerator employing a hardware/software co-design that supports multiple parameter sets (Kyber512, 768, and 1024). Implemented on an FPGA-based System-on-Chip (SoC), their design achieved a fourfold increase in throughput compared to software-only implementations while maintaining flexibility for parameter switching and protocol adaptation. This study demonstrated that combining embedded processors with reconfigurable logic offers a balanced trade-off between performance and adaptability in IoT contexts.

In a complementary line of research, Kieu-Do-Nguyen et al. (2024) introduced a compact and low-latency Number Theoretic Transform (NTT) architecture for Kyber, significantly reducing computation delay in key generation and encapsulation phases. Their FPGA implementation optimized pipelining and memory access, cutting latency by approximately 30 % relative to conventional NTT architectures. Similarly, Dolmeta et al. (2023) designed a Keccak accelerator on RISC-V processors to enhance Kyber's hashing and message-digest operations, achieving a threefold improvement in hashing speed with minimal energy overhead.

Other studies, including those by Ye et al. (2024) and Cíbik et al. (2025), validated that hardware-assisted Kyber implementations can outperform classical elliptic-curve cryptography (ECC) both in speed and energy efficiency when optimized for edge-level devices. These results underline a growing consensus that FPGA- and RISC-V-based PQC architectures are viable for future IoT systems that require secure yet low-power cryptographic computation.

Despite these promising results, hardware-level PQC faces ongoing challenges, particularly regarding scalability, manufacturing cost, and side-channel resistance. Many prototypes remain confined to laboratory environments and require further testing under realistic IoT workloads. Nonetheless, these developments provide strong evidence that co-designed PQC architectures can bridge the gap between high security and energy-aware efficiency, paving the way toward practical post-quantum deployment in embedded systems.

E. Comparative Performance and Trade-Offs

A comparative synthesis of the fifteen selected studies highlights significant progress in adapting post-quantum cryptography for IoT systems, particularly through the implementation and optimization of the CRYSTALS-Kyber algorithm. Across the reviewed works, researchers consistently focus on addressing the core challenge of balancing security, computational efficiency, and energy consumption within constrained IoT architectures. The findings indicate that while Kyber offers robust quantum-resistant protection through its lattice-based foundation, its computational intensity requires various levels of algorithmic, hardware, and protocol optimizations to achieve practical deployment in real-world IoT networks.

Most studies converge on the conclusion that Kyber512 provides the optimal compromise between performance and cryptographic strength. Mahdi and Abdullah (2025) identified Kyber512 as the most energy-efficient variant, meeting NIST's Level 1 security target while minimizing memory use. Similarly, Fitzgibbon and Ottaviani (2024) benchmarked Kyber against other NIST candidates such as Saber and NTRU, finding Kyber to deliver faster key generation and decapsulation times on embedded microcontrollers. Yet, increasing to Kyber768 or Kyber1024 incurs up to triple the computational cost, making them impractical for ultra-low-power devices. Further arithmetic-level optimization by Huang et al. (2024) and Ye et al. (2024)—including refined modular reduction and FPGA-based lattice processors—demonstrated notable reductions in latency and processing time. These enhancements validate that Kyber can indeed be adapted into a lightweight yet quantum-secure cryptosystem suitable for edge and sensor nodes.

At the protocol level, several studies successfully extended these improvements into communication

frameworks. Kim and Seo (2025) developed KEM-MQTT, a Kyber-based variant of the MQTT protocol optimized for 8-bit AVR microcontrollers, achieving up to 81% faster key operations while maintaining compatibility with TLS 1.3. Meanwhile, Torre et al. (2025) proposed Wireless Kyber (WKyber), which embeds Kyber at the physical layer using Additive White Gaussian Noise (AWGN) and BCH coding to transmit coefficients securely. Both approaches demonstrate that protocol-level integration can provide quantum-safe communication with minimal redesign of existing IoT infrastructure, bridging classical and post-quantum systems efficiently.

Despite these advancements, trade-offs persist. Hardware-based acceleration yields higher throughput and lower latency but increases power consumption and design complexity, as shown by Dinh et al. (2024). Conversely, software-only implementations remain portable yet struggle to maintain performance at higher security levels. Hybrid cryptographic frameworks combining Kyber with ECC or TPM-based methods enhance interoperability but lengthen handshake processes and enlarge payload sizes. Overall, the comparative analysis reveals a clear trend toward multi-dimensional optimization, where algorithmic efficiency, hardware acceleration, and communication integration must be jointly addressed. Among the available configurations, Kyber512 remains the most practical for IoT environments, while higher-level variants are better suited for gateways or industrial-grade systems. Future designs must therefore aim for a unified adaptive encryption model—capable of dynamically adjusting cryptographic parameters and balancing energy efficiency, throughput, and interoperability—to achieve scalable and sustainable post-quantum IoT security.

F. Identified Gaps and Future Directions

Although significant advancements have been made in adapting CRYSTALS-Kyber for post-quantum IoT security, the collective findings of the fifteen reviewed studies reveal that several key challenges remain unresolved. These challenges primarily concern the integration, scalability, and energy efficiency of post-quantum cryptographic systems when deployed in large-scale, heterogeneous IoT environments.

The first and most fundamental gap lies in the lack of a unified lightweight framework that integrates algorithmic optimization, communication protocol adaptation, and hardware acceleration within a single architecture. Most existing works focus on improving one layer of the cryptographic stack in isolation—such as arithmetic computation, handshake protocol, or FPGA acceleration—without providing a holistic model that allows these components to operate synergistically. This limitation often results in performance fragmentation, where gains achieved in one area lead to inefficiencies or incompatibilities in another.

Secondly, there is a persistent energy-performance trade-off across most hardware-based and hybrid implementations. Studies such as those by Dinh et al. (2024) and Ye et al. (2024) achieved impressive throughput gains through FPGA and co-design approaches, yet these improvements often come with increased power consumption, which is unsustainable for battery-powered IoT devices. Similarly, protocol-level schemes such as KEM-MQTT and Wireless Kyber reduce computational overhead but have not yet been extensively validated under real deployment scenarios involving variable network conditions, interference, and node mobility. This indicates a need for energy-aware cryptographic frameworks capable of dynamically adjusting encryption parameters based on device capability and environmental context.

Another critical gap concerns standardization and interoperability. While NIST's finalization of ML-KEM (Kyber) in 2024 provides a stable foundation for post-quantum security, the adaptation of this standard to IoT-specific protocols such as MQTT, CoAP, and OPC-UA remains limited. Most existing implementations rely on custom modifications that hinder cross-platform compatibility. Without a standardized interface for post-quantum key exchange, large-scale IoT ecosystems risk fragmentation, reducing the reliability and maintainability of security infrastructures across heterogeneous hardware and communication layers.

Finally, the reviewed works reveal a shortage of comprehensive security evaluations against side-channel attacks and physical vulnerabilities. As IoT devices are often deployed in uncontrolled environments, their cryptographic modules are highly susceptible to timing, power, and electromagnetic leakage attacks. Only a few studies, such as Hoang et al. (2024), address side-channel resistance in their design, leaving a critical gap in the practical robustness of lightweight post-quantum implementations.

Looking ahead, future research should focus on developing a Lightweight Adaptive Post-Quantum Encryption Framework that seamlessly combines software and hardware optimizations, supports energy-aware computation, and conforms to IoT communication standards. Such a framework should employ modular design principles, enabling flexible integration across diverse hardware platforms—from 8-bit sensor nodes to edge servers—while maintaining scalable post-quantum security. Furthermore, experimental validation on real-world IoT testbeds, such as ESP32 or Raspberry Pi, will be essential for assessing the practicality of these models under realistic latency, energy, and connectivity constraints.

In addition, future work should explore hybrid integration with AI-driven optimization, where adaptive machine learning models dynamically tune Kyber parameters to minimize power consumption while preserving encryption strength. This cross-disciplinary direction can bridge the gap between post-quantum cryptography and intelligent IoT security, enabling self-optimizing, context-aware encryption in large-scale IoT deployments.

Moreover, collaboration between academia, industry, and standardization bodies such as NIST, ISO, and ETSI will be essential to ensure the alignment of post-quantum cryptography standards with real-world IoT requirements. Establishing shared benchmarks, open-source implementations, and testbeds for Kyber-based IoT encryption will accelerate validation, interoperability, and global adoption. Such coordinated efforts will transform post-quantum cryptography from experimental prototypes

into dependable, scalable infrastructures for securing the rapidly expanding Internet of Things ecosystem.

IV. CONCLUSION

This study conducted a comprehensive literature review and analysis of fifteen selected works related to the implementation of CRYSTALS-Kyber as a lightweight post-quantum cryptographic solution for the Internet of Things (IoT). The findings indicate that Kyber, as a lattice-based algorithm standardized by NIST in 2024, provides strong resistance against quantum attacks and has shown promising adaptability to constrained IoT environments. Through an in-depth comparison of existing research, three dominant approaches were identified—algorithmic optimization, protocol-level integration, and hardware co-design—each contributing uniquely to the advancement of post-quantum security in resource-limited systems.

The review reveals that Kyber512 is the most practical variant for lightweight IoT deployment, balancing computational efficiency, key size, and cryptographic robustness. Algorithmic improvements such as modular arithmetic optimization and polynomial compression significantly reduce computational overhead, while FPGA and RISC-V-based hardware implementations deliver substantial performance gains for edge computing applications. Moreover, integrating Kyber into communication protocols like MQTT and TLS enables quantum-safe key exchange with minimal latency, proving that post-quantum cryptography can coexist with existing IoT infrastructures.

However, several challenges persist. Current implementations often optimize individual layers—algorithmic, hardware, or protocol—without providing a unified design that harmonizes efficiency, scalability, and interoperability. Energy consumption remains a critical concern, particularly in hardware-accelerated or hybrid frameworks, and standardization across IoT communication protocols is still limited. Addressing these limitations will require the development of an adaptive, modular post-quantum encryption framework capable of dynamically tuning its cryptographic parameters based on device resources, network conditions, and application requirements.

In conclusion, the Lightweight Post-Quantum Encryption Model using CRYSTALS-Kyber represents a promising pathway toward securing next-generation IoT systems against emerging quantum threats. By combining algorithmic refinement, protocol integration, and energy-aware hardware co-design, future research can establish scalable and sustainable post-quantum security foundations. Such advancements will not only safeguard IoT ecosystems but also ensure that quantum-resilient communication becomes an integral component of digital transformation in the post-quantum era.

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